

Milestone 4: Statistical Inference & Analytical Modeling

E-commerce Customer Behavior Analytics System

1. Executive Summary

Building upon the foundational analysis and visualizations, Milestone 4 implements advanced analytical modeling and statistical inference. This phase tests the ability to extract meaningful insights through mathematical rigor, specifically focusing on predictive modeling for customer spending and feature impact analysis.

2. Analytical Modeling Approach

To support decision-making, a **Linear Regression Model** was developed to predict 'Total Spend' based on customer demographics and behavioral metrics.

Model Implementation Details:

- **Feature Selection:** Targeted inputs included Age, Items Purchased, and Satisfaction Scores, while removing non-predictive identifiers like 'Customer ID'.
- **Data Partitioning:** Implemented a 80/20 Train-Test split to ensure model generalizability.
- **Encoding:** Applied dynamic One-Hot Encoding to handle remaining categorical strings.

3. Statistical Evaluation & Results

The model performance was evaluated using standard statistical metrics to interpret uncertainty and predictive accuracy.

Metric	Interpretation
R-Squared (R^2)	Measures the proportion of variance in spending explained by the model features.
Mean Squared Error (MSE)	Quantifies the average squared difference between estimated and actual spending values.
Feature Coefficients	Statistically ranks features by their weight in influencing the target variable (Total Spend).

4. Inference & Insight Validation

The analytical modeling phase confirmed critical behavioral trends:

- **Positive Correlations:** 'Items Purchased' showed a dominant positive coefficient, validating that volume remains the primary driver of revenue.
- **Satisfaction Impact:** The model quantified the monetary value of moving a customer from 'Unsatisfied' to 'Satisfied'.
- **Demographic Influence:** Specific cities or membership tiers showed statistically significant shifts in expected total spend.

5. Interpretation of Uncertainty

By analyzing the MSE and residuals, the system identifies the variability in customer behavior. This allows for the interpretation of "high-risk" predictions versus standard behavioral patterns, fulfilling the requirement for interpreting uncertainty and variability in analytical systems.